

Conversational Language Understanding – Code Snippets

■ Python RESTful API Example

```
import requests, json

url = f"{endpoint}/language/:analyze-conversations?projectName=YOUR_CLU_PROJECT&deploymentName=production"
headers = {
    "Ocp-Apim-Subscription-Key": key,
    "Content-Type": "application/json"
}
body = {
    "kind": "Conversation",
    "analysisInput": {
        "conversationItem": {"participantId": "1", "text": "Book a flight to Seattle"}
    }
}

def main():
    res = requests.post(url, headers=headers, json=body)
    print(json.dumps(res.json(), indent=2))

if __name__ == "__main__":
    main()
```

■ Python OLD SDK (LUIS Runtime)

```
# pip install azure-cognitiveservices-language-luis
from azure.cognitiveservices.language.luis.runtime import LUISRuntimeClient
from msrest.authentication import CognitiveServicesCredentials

def main():
    endpoint = "YOUR_LUIS_ENDPOINT"
    key = "YOUR_KEY"
    app_id = "YOUR_APP_ID"
    slot = "production"

    client = LUISRuntimeClient(endpoint, CognitiveServicesCredentials(key))
    prediction = client.prediction.get_slot_prediction(app_id, slot, {"query": "Book a flight to Seattle"})
    print(prediction.prediction.top_intent)
    print(prediction.prediction.entities)

if __name__ == "__main__":
    main()
```

■ Python NEW SDK (Azure.AI.Language.Conversations)

```
# pip install azure-ai-language-conversations
from azure.ai.language.conversations import ConversationAnalysisClient
from azure.core.credentials import AzureKeyCredential

def main():
    endpoint = "YOUR_LANGUAGE_ENDPOINT"
    key = "YOUR_KEY"
    project = "YOUR_CLU_PROJECT"
    deployment = "production"
    client = ConversationAnalysisClient(endpoint, AzureKeyCredential(key))
```

```

    result = client.analyze_conversation(
        task={
            "kind": "Conversation",
            "analysisInput": {"conversationItem": {"participantId": "1", "text": "Book a flight to Seat
            "parameters": {"projectName": project, "deploymentName": deployment}
        }
    )
    print(result["result"]["prediction"]["topIntent"])

if __name__ == "__main__":
    main()

```

■ C# RESTful API Example

```

using System;
using System.Net.Http;
using System.Text;

class Program
{
    static void Main()
    {
        var endpoint = "YOUR_ENDPOINT";
        var key = "YOUR_KEY";

        var client = new HttpClient();
        client.DefaultRequestHeaders.Add("Ocp-Apim-Subscription-Key", key);

        var body = new {
            kind = "Conversation",
            analysisInput = new { conversationItem = new { participantId = "1", text = "Book a flight to
        };
        var json = System.Text.Json.JsonSerializer.Serialize(body);

        var res = client.PostAsync(
            $"{endpoint}/language/:analyze-conversations?projectName=YOUR_CLU_PROJECT&deploymentName=pr
            new StringContent(json, Encoding.UTF8, "application/json")
        ).GetAwaiter().GetResult();

        Console.WriteLine(res.Content.ReadAsStringAsync().GetAwaiter().GetResult());
    }
}

```

■ C# OLD SDK (LUIS Runtime)

```

// dotnet add package Microsoft.Azure.CognitiveServices.Language.LUIS.Runtime
using System;
using Microsoft.Azure.CognitiveServices.Language.LUIS.Runtime;
using Microsoft.Azure.CognitiveServices.Language.LUIS.Runtime.Models;

class Program
{
    static void Main()
    {
        string endpoint = "YOUR_LUIS_ENDPOINT";
        string key = "YOUR_KEY";
        Guid appId = Guid.Parse("YOUR_APP_ID");
        string slot = "production";
        var client = new LUISRuntimeClient(new ApiKeyServiceClientCredentials(key)) { Endpoint = endpoi
    }
}

```

```

        var response = client.Prediction.GetSlotPredictionAsync(appId, slot, new PredictionRequest { Question = question });
        response
            .GetAwaiter().GetResult();
        Console.WriteLine(response.Prediction.TopIntent);
    }
}

```

■ C# NEW SDK (Azure.AI.Language.Conversations)

```

// dotnet add package Azure.AI.Language.Conversations
using System;
using Azure;
using Azure.AI.Language.Conversations;

class Program
{
    static void Main()
    {
        string endpoint = "YOUR_LANGUAGE_ENDPOINT";
        string key = "YOUR_KEY";
        string project = "YOUR_CLU_PROJECT";
        string deployment = "production";

        var client = new ConversationAnalysisClient(new Uri(endpoint), new AzureKeyCredential(key));
        var data = new ConversationAnalysisData
        {
            kind = "Conversation",
            analysisInput = new { conversationItem = new { participantId = "1", text = "Book a flight to New York" },
            parameters = new { projectName = project, deploymentName = deployment }
        };
        var json = System.Text.Json.JsonSerializer.Serialize(data);
        var result = client.AnalyzeConversation(BinaryData.FromString(json));
        Console.WriteLine(result.ToString());
    }
}

```